THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 5: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest 3-digit palindrome divisible by 4 is: 1

(a) 202 **(b)** 216 **(c)** 222 **(d)** 212

2. If the number 1a4b is divisible by 90, then $a + b$ equals: 1

(a) 5 **(b)** 4 **(c)** 7 **(d)** 6

3. The number of steps the Kaprekar routine needs to reach 6174 starting from 5554 is: 1

(a) 5 **(b)** 7 **(c)** 4 **(d)** 6

4. A number is divisible by both 8 and 9. It must be divisible by: 1

(a) 72 **(b)** 17 **(c)** 80 **(d)** 144

5. The smallest number which leaves remainder 4 when divided by 14 and 21 is: 1

(a) 88 **(b)** 42 **(c)** 46 **(d)** 298

6. The number of multiples of 15 between 50 and 1000 (both included) is: 1

(a) 64 **(b)** 65 **(c)** 62 **(d)** 63

7. The number of 3-digit numbers divisible by 17 is: 1

(a) 54 **(b)** 53 **(c)** 52 **(d)** 58

8. The smallest 3-digit palindrome divisible by 6 is: 1

(a) 232 **(b)** 222 **(c)** 228 **(d)** 212

9. **Assertion (A):** A number divisible by both 2 and 3 is divisible by 6. 1
Reason (R): A number is divisible by 11 if the sum of its digits is divisible by 11.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The number 2,345 is divisible by 9. 1
Reason (R): A number is divisible by 9 if and only if the sum of its digits is divisible by 9.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How many multiples of 14 lie between 250 and 1500 (both included)? 2

12. The sum of five consecutive numbers is 80. Find the numbers. 2

13. Find the smallest number which leaves remainder 9 when divided by 28 and 42. 2

14. How many multiples of 6 lie between 75 and 600 (both included)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the six-digit number 429429 is divisible by 7, 11 and 13. 3

16. Find the greatest 4-digit number divisible by 9, 15 and 18. 3

17. How many 3-digit numbers are divisible by both 10 and 14? 3

OR

Show that the six-digit number 516516 is divisible by 7, 11 and 13. 3

18. Find the greatest 4-digit number divisible by 14, 21 and 28. 3

OR

How many 3-digit numbers are divisible by both 15 and 20? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Riya applies Kaprekar's routine to numbers: arrange the digits to form the largest and the smallest numbers and subtract. She starts with the three-digit number 352 and then tries the four-digit number 4312. 5

(i) Find the first step for 352.
(ii) Find the second step for 352.
(iii) Which number does 352 finally reach?
(iv) Apply the routine to 4312 and find the number reached and how many steps are needed.

20. Find the digits a and b if the number 8a1b is divisible by 45. List all possibilities. 5

OR

Take the number 723. Subtract its reverse. Add the reverse of the result to the result. What do you get? Show that this always happens for 3-digit numbers whose first and last digits differ by 2 or more. 5

END OF QUESTION PAPER — Total: 40 Marks

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